

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION — Paper 3 —
SOLUTIONS AND MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “*(From formula sheet)*” indicates the equation is on the provided formula sheet.
 - “*(Recalled — not on formula sheet)*” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
-

Section A: Oscillations and Waves — Answers

1. C — The period of a mass-spring system is $T = 2\pi\sqrt{m/k}$ (*From formula sheet*). Doubling the mass gives $T' = 2\pi\sqrt{2m/k} = \sqrt{2}T$. The period increases by a factor of $\sqrt{2}$.

2. C — In SHM, the total energy is $E = \frac{1}{2}kA^2$. At displacement x : $\text{PE} = \frac{1}{2}kx^2$, $\text{KE} = \frac{1}{2}k(A^2 - x^2)$. Setting $\text{KE} = 3\text{PE}$: $A^2 - x^2 = 3x^2$, so $x^2 = A^2/4$, giving $x = A/2$.

3. C — The amplitude decays as $A(t) = A_0e^{-\gamma t/2}$ (*From formula sheet*). It reaches A_0/e when $\gamma t/2 = 1$, i.e. $t = 2/\gamma$. The number of cycles is $n = t/T \approx \omega_0/(2\pi\gamma) = Q/\pi$. For $Q = 50$: $n = 50/\pi \approx 16$ cycles.

4. B — The phase lag satisfies $\tan \delta = \gamma\omega/(\omega_0^2 - \omega^2)$ (*From formula sheet*). For $\delta = \pi/2$, $\tan \delta \rightarrow \infty$, which requires the denominator $\omega_0^2 - \omega^2 = 0$, i.e. $\omega = \omega_0$.

5. C — At a fixed (rigid) end, the displacement must be zero. This boundary condition is satisfied by a reflected pulse that is inverted (phase change of π). This can also be seen from the reflection coefficient $r = (Z_1 - Z_2)/(Z_1 + Z_2)$ (*From formula sheet*) with $Z_2 \rightarrow \infty$, giving $r \rightarrow -1$.

6. B — The string impedance is $Z = \sqrt{\mu T}$ (*From formula sheet*). $Z = \sqrt{0.04 \times 100} = \sqrt{4} = 2.0 \text{ kg/s}$.

7. B — The sound level in decibels is $\beta = 10 \log_{10}(I/I_0)$ (*From formula sheet*). If I doubles: $\Delta\beta = 10 \log_{10}(2I/I_0) - 10 \log_{10}(I/I_0) = 10 \log_{10}(2) = 10 \times 0.301 = 3.01 \text{ dB} \approx 3.0 \text{ dB}$.

8. A — The Doppler formula is $f' = f \frac{v + v_o}{v - v_s}$ (*From formula sheet*) (upper signs for approach). $f' = 500 \times \frac{340 + 10}{340 - 20} = 500 \times \frac{350}{320} = 500 \times 1.09375 = 546.9 \text{ Hz} \approx 547 \text{ Hz}$.

9. C — In the n th harmonic on a string fixed at both ends, the number of nodes is $n + 1$ (including both endpoints). For $n = 4$: there are $4 + 1 = 5$ nodes.

10. A — For two identical pendulums coupled by a spring: In the symmetric (in-phase) mode, both pendulums swing together and the coupling spring is never stretched, so $\omega_1 = \sqrt{g/L}$, i.e. $f_1 = \sqrt{g/L}/(2\pi)$. In the antisymmetric (antiphase) mode, the spring is alternately stretched and compressed, contributing an additional restoring force $2k_c x$ to each pendulum, so $\omega_2 = \sqrt{g/L + 2k_c/m}$, i.e. $f_2 = \sqrt{g/L + 2k_c/m}/(2\pi)$.

Section B: Quantum Physics — Answers

1. **D** — $\langle a|b\rangle$ is a scalar (complex number) and $\langle c|d\rangle$ is also a scalar. The product of two scalars is a scalar.
2. **C** — A valid quantum state must be normalised: $\langle\psi|\psi\rangle = 1$. For option C: $|1/\sqrt{2}|^2 + |i/\sqrt{2}|^2 = 1/2 + 1/2 = 1$. Option A gives norm $\sqrt{2}$; option B gives $|1/2|^2 + |1/2|^2 = 1/4 + 1/4 = 1/2 \neq 1$; option D gives $|1/\sqrt{2}|^2 + |1|^2 = 1/2 + 1 = 3/2 \neq 1$.
3. **C** — The completeness relation states that $\sum_n |n\rangle\langle n| = \hat{I}$ (the identity operator) for any complete orthonormal basis. (*Recalled — not on formula sheet*)
4. **B** — The Hermitian conjugate of a product reverses the order: $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$. (*Recalled — not on formula sheet*)
5. **B** — The eigenvalues satisfy $\det(M - \lambda I) = 0$: $(3-\lambda)(0-\lambda)-4 = 0$, i.e. $-3\lambda + \lambda^2 - 4 = 0$, so $\lambda^2 - 3\lambda - 4 = 0$, which factors as $(\lambda-4)(\lambda+1) = 0$. Eigenvalues: $\lambda = 4$ and $\lambda = -1$.
6. **C** — Upon measuring observable A and obtaining eigenvalue a_n (non-degenerate), the state collapses to the corresponding eigenstate $|a_n\rangle$. This is the measurement postulate (Born rule/projection postulate). (*Recalled — not on formula sheet*)
7. **C** — The time evolution of an energy eigenstate is $|\psi(t)\rangle = e^{-iE_1 t/\hbar} |E_1\rangle$ (*From formula sheet*). Note the minus sign in the exponent.
8. **C** — For spin-1/2: $s = 1/2$. The eigenvalue of $\hat{S}^2$ is $s(s+1)\hbar^2 = \frac{1}{2} \cdot \frac{3}{2} \hbar^2 = \frac{3}{4} \hbar^2$. Both spin-up and spin-down states share this same eigenvalue of $\hat{S}^2$.
9. **A** — Two identical fermions must be in an antisymmetric state under exchange. The only antisymmetric combination of two states $|\alpha\rangle$ and $|\beta\rangle$ is $\frac{1}{\sqrt{2}}(|\alpha\rangle|\beta\rangle - |\beta\rangle|\alpha\rangle)$. There is only 1 distinct two-particle state. (Both particles in the same state, e.g. $|\alpha\rangle|\alpha\rangle$, is forbidden by the Pauli exclusion principle.)
10. **B** — The proton is composed of two up quarks and one down quark: uud . This gives charge $+\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$, consistent with the proton charge. (*Recalled — not on formula sheet*)

Section C: Long Answer Solutions

Question 1.

[15 marks]

Collisions, Impulse, and Centre of Mass

- (a) Take the positive direction as the direction of the 4 kg trolley's motion (to the right).

Initial momenta: (*Recalled — not on formula sheet*)

$$p_1 = m_1 v_1 = 4 \times 6 = 24 \text{ kg m/s} \quad [1]$$

$$p_2 = m_2 v_2 = 2 \times (-3) = -6 \text{ kg m/s}$$

$$\text{Total initial momentum: } p_{\text{total}} = 24 + (-6) = 18 \text{ kg m/s} \quad [1]$$

In a perfectly inelastic collision, the two objects stick together. By conservation of momentum:

$$(m_1 + m_2) v_f = p_{\text{total}}$$

$$6 v_f = 18$$

$$v_f = 3 \text{ m/s to the right (in the direction of the 4 kg trolley).}$$

[1]

- (b) Kinetic energy before the collision: (*Recalled — not on formula sheet*)

$$\text{KE}_{\text{before}} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} (4)(6^2) + \frac{1}{2} (2)(3^2) = 72 + 9 = 81 \text{ J} \quad [1]$$

Kinetic energy after the collision:

$$\text{KE}_{\text{after}} = \frac{1}{2} (m_1 + m_2) v_f^2 = \frac{1}{2} (6)(3^2) = 27 \text{ J} \quad [1]$$

Fraction of initial KE lost:

$$\frac{\text{KE}_{\text{before}} - \text{KE}_{\text{after}}}{\text{KE}_{\text{before}}} = \frac{81 - 27}{81} = \frac{54}{81} = \frac{2}{3} \approx 66.7\%$$

[1]

- (c) Using the elastic collision formula for v'_1 from the formula sheet: (*From formula sheet*)

$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

[1]

Substituting values:

$$v'_1 = \frac{4 - 2}{4 + 2} \times 6 + \frac{2 \times 2}{4 + 2} \times (-3) = \frac{2}{6} \times 6 + \frac{4}{6} \times (-3) = 2 + (-2) = 0 \text{ m/s}$$

[1]

For v'_2 , we can use conservation of momentum (or the analogous formula with indices swapped):

$$m_1 v_1 + m_2 v_2 = m_1 v'_1 + m_2 v'_2 \quad [1]$$

$$18 = 4(0) + 2 v'_2$$

$$v'_2 = 9 \text{ m/s}$$

$$\text{Verification: KE after} = \frac{1}{2}(4)(0)^2 + \frac{1}{2}(2)(9)^2 = 0 + 81 = 81 \text{ J} = \text{KE}_{\text{before}} \checkmark$$

$$v'_1 = 0 \text{ m/s (the 4 kg trolley stops); } v'_2 = 9 \text{ m/s to the right.}$$

[1]

(d) The impulse is the change in momentum: (*Recalled — not on formula sheet*)

$$J = \Delta p = m(v_f - v_i) = 3 \times (8 - 2) = 18 \text{ N s} \quad [1]$$

The average force is:

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{18}{0.5} = 36 \text{ N}$$

[1]

(e) The centre of mass position is: (*From formula sheet*)

$$\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i} \quad [1]$$

$$x_{\text{cm}} = \frac{1 \times 0 + 2 \times 3 + 3 \times 0}{1 + 2 + 3} = \frac{0 + 6 + 0}{6} = 1 \text{ m} \quad [1]$$

$$y_{\text{cm}} = \frac{1 \times 0 + 2 \times 0 + 3 \times 4}{6} = \frac{0 + 0 + 12}{6} = 2 \text{ m}$$

$$\mathbf{R}_{\text{cm}} = (1, 2) \text{ m}$$

[1]

Question 2.

[15 marks]

Thermodynamic Cycle

- (a) Using the ideal gas law: (
- From formula sheet*
-)

$$PV = nRT$$

[1]

$$V_A = \frac{nRT_A}{P_A} = \frac{2 \times 8.314 \times 300}{1.00 \times 10^5}$$

[1]

$$V_A = \frac{4988.4}{1.00 \times 10^5}$$

$$V_A = 0.0499 \text{ m}^3 \approx 49.9 \text{ L}$$

[1]

- (b) (i) At constant pressure,
- $V \propto T$
- (from
- $PV = nRT$
-).

[1]

$$V_B = V_A \times \frac{T_B}{T_A} = 0.0499 \times \frac{600}{300} = 0.0998 \text{ m}^3$$

- (ii) Work done by the gas in an isobaric process: (
- Recalled — not on formula sheet*
-)

$$W = P_A(V_B - V_A) = 1.00 \times 10^5 \times (0.0998 - 0.0499) = 1.00 \times 10^5 \times 0.0499$$

[1]

$$W_{A \rightarrow B} = 4990 \text{ J} \approx 4.99 \text{ kJ}$$

- (iii) Heat absorbed at constant pressure: (
- Recalled — not on formula sheet*
-)

$$Q = nC_P \Delta T$$

For a monatomic ideal gas, $C_P = \frac{5}{2}R$ (since $C_V = \frac{3}{2}R$ and $C_P - C_V = R$). (*Recalled — not on formula sheet*)

[1]

$$Q = 2 \times \frac{5}{2} \times 8.314 \times (600 - 300) = 2 \times 20.785 \times 300$$

$$Q_{A \rightarrow B} = 12,471 \text{ J} \approx 12.5 \text{ kJ}$$

[1]

Check: $\Delta U = nC_V \Delta T = 2 \times \frac{3}{2} \times 8.314 \times 300 = 7483 \text{ J}$. First law: $\Delta U = Q - W = 12471 - 4990 = 7481 \text{ J}$ ✓

- (c) For an adiabatic process:
- $TV^{\gamma-1} = \text{const}$
- (
- From formula sheet*
-)

$$T_B V_B^{\gamma-1} = T_C V_C^{\gamma-1}$$

[1]

With $\gamma - 1 = 2/3$, $T_B = 600 \text{ K}$, $T_C = 300 \text{ K}$:

$$V_C^{2/3} = V_B^{2/3} \times \frac{T_B}{T_C} = (0.0998)^{2/3} \times \frac{600}{300} = (0.0998)^{2/3} \times 2$$

[1]

Computing: $(0.0998)^{2/3} = e^{(2/3) \ln(0.0998)} = e^{(2/3)(-2.306)} = e^{-1.537} = 0.2154$

$$V_C^{2/3} = 0.2154 \times 2 = 0.4309$$

$$V_C = (0.4309)^{3/2} = 0.4309 \times \sqrt{0.4309} = 0.4309 \times 0.6564 = 0.2828 \text{ m}^3$$

$$V_C \approx 0.283 \text{ m}^3$$

[1]

Using the ideal gas law for the pressure:

$$P_C = \frac{nRT_C}{V_C} = \frac{2 \times 8.314 \times 300}{0.2828} = \frac{4988.4}{0.2828}$$

$$P_C \approx 1.76 \times 10^4 \text{ Pa}$$

[1]

Verification: $P_B V_B^\gamma = 1.00 \times 10^5 \times (0.0998)^{5/3} = 1.00 \times 10^5 \times 0.02148 = 2148$.
 $P_C V_C^\gamma = 1.76 \times 10^4 \times (0.2828)^{5/3} = 1.76 \times 10^4 \times 0.1218 = 2144$. ✓ (small rounding error).

- (d) (i) Isothermal compression from $V_C = 0.283 \text{ m}^3$ to $V_A = 0.0499 \text{ m}^3$ at $T = 300 \text{ K}$:

$$W_{\text{by gas}} = nRT \ln\left(\frac{V_A}{V_C}\right) = 2 \times 8.314 \times 300 \times \ln\left(\frac{0.0499}{0.283}\right) \text{ (From formula sheet)}$$

$$= 4988.4 \times \ln(0.1764) = 4988.4 \times (-1.735)$$

[1]

$$W_{\text{by gas}} = -8655 \text{ J}$$

Work done *on* the gas = $+8655 \text{ J} \approx 8.66 \text{ kJ}$.

- (ii) Total work done by the gas in one complete cycle:

Step A→B (isobaric): $W_1 = 4990 \text{ J}$

Step B→C (adiabatic): $W_2 = -\Delta U = nC_V(T_B - T_C) = 2 \times \frac{3}{2} \times 8.314 \times 300 = 7483 \text{ J}$

Step C→A (isothermal): $W_3 = -8655 \text{ J}$

$$W_{\text{cycle}} = W_1 + W_2 + W_3 = 4990 + 7483 - 8655 = 3818 \text{ J} \approx 3.82 \text{ kJ}$$

- (iii) The heat *input* occurs only during the isobaric heating step (A→B), where $Q_{\text{in}} = 12,471 \text{ J}$.

(In the adiabatic step $Q = 0$; in the isothermal compression, heat is *released*: $Q_{C \rightarrow A} = W_3 = -8655 \text{ J}$.)

[1]

Efficiency:

$$\eta = \frac{W_{\text{cycle}}}{Q_{\text{in}}} = \frac{3818}{12471} = 0.306 = 30.6\%$$

Carnot efficiency between $T_c = 300 \text{ K}$ and $T_h = 600 \text{ K}$: (From formula sheet)

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{600} = 0.5 = 50\%$$

The cycle efficiency is $\eta \approx 30.6\%$, which is less than the Carnot efficiency $\eta_C = 50\%$, as required by the second law of thermodynamics.

[1]

Question 3.

[15 marks]

Coupled Oscillators and Pendulums

- (a) Mass 1 is attached to two springs: one connected to the wall (restoring force $-kx_1$) and one connected to mass 2 (restoring force $-k(x_1 - x_2)$). [1]

$$m\ddot{x}_1 = -kx_1 - k(x_1 - x_2) = -2kx_1 + kx_2$$

Similarly, mass 2 is attached to a spring connected to mass 1 and a spring connected to the wall: [1]

$$m\ddot{x}_2 = -k(x_2 - x_1) - kx_2 = kx_1 - 2kx_2$$

$$0.5 \ddot{x}_1 = -80 x_1 + 40 x_2$$

$$0.5 \ddot{x}_2 = 40 x_1 - 80 x_2$$

Or equivalently: $\ddot{x}_1 = -160 x_1 + 80 x_2$ and $\ddot{x}_2 = 80 x_1 - 160 x_2$. [1]

- (b) Seek solutions $x_1 = A_1 \cos(\omega t)$, $x_2 = A_2 \cos(\omega t)$. Substituting: [1]

$$(2k - m\omega^2)A_1 - kA_2 = 0$$

$$-kA_1 + (2k - m\omega^2)A_2 = 0$$

For non-trivial solutions, the determinant must vanish: [1]

$$(2k - m\omega^2)^2 - k^2 = 0$$

$$2k - m\omega^2 = \pm k$$

Mode 1 ($+k$): $m\omega_1^2 = k$, so $\omega_1^2 = k/m = 40/0.5 = 80 \text{ rad}^2/\text{s}^2$

$$\omega_1 = \sqrt{80} = 4\sqrt{5} \approx 8.94 \text{ rad/s}$$

In this mode, $A_1 = A_2$: both masses move **in phase** (symmetric mode). [1]

Mode 2 ($-k$): $m\omega_2^2 = 3k$, so $\omega_2^2 = 3k/m = 120 \text{ rad}^2/\text{s}^2$

$$\omega_2 = \sqrt{120} = 2\sqrt{30} \approx 10.95 \text{ rad/s}$$

In this mode, $A_1 = -A_2$: the masses move in **antiphase** (antisymmetric mode). [1]

- (c) Define normal coordinates: $q_1 = x_1 + x_2$ (in-phase) and $q_2 = x_1 - x_2$ (antiphase).

Then $x_1 = \frac{1}{2}(q_1 + q_2)$ and $x_2 = \frac{1}{2}(q_1 - q_2)$.

Initial conditions: $x_1(0) = 0.05 \text{ m}$, $x_2(0) = 0$, $\dot{x}_1(0) = \dot{x}_2(0) = 0$. [1]

So: $q_1(0) = 0.05 + 0 = 0.05 \text{ m}$, $q_2(0) = 0.05 - 0 = 0.05 \text{ m}$, $\dot{q}_1(0) = \dot{q}_2(0) = 0$.

Each normal coordinate oscillates at its own frequency with zero initial velocity, so:

$$q_1(t) = 0.05 \cos(\omega_1 t), \quad q_2(t) = 0.05 \cos(\omega_2 t) \quad [1]$$

Therefore:

$$x_1(t) = 0.025 [\cos(\omega_1 t) + \cos(\omega_2 t)]$$

$$x_2(t) = 0.025 [\cos(\omega_1 t) - \cos(\omega_2 t)]$$

where $\omega_1 = 4\sqrt{5} \approx 8.94$ rad/s and $\omega_2 = 2\sqrt{30} \approx 10.95$ rad/s. [1]

This describes beats: energy transfers periodically between the two masses at the beat frequency $|\omega_2 - \omega_1|/2$.

(d) Using the simple pendulum formula: (*From formula sheet*)

$$T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{1.2}{9.81}} \quad [1]$$

$$= 2\pi\sqrt{0.1223} = 2\pi \times 0.3497$$

$$T = 2.20 \text{ s}$$

[1]

(e) (i) Moment of inertia of a uniform thin rod about one end: (*From formula sheet*)

$$I = \frac{1}{3}M\ell^2 = \frac{1}{3} \times 0.8 \times (0.6)^2 = \frac{1}{3} \times 0.8 \times 0.36$$

$$I = 0.096 \text{ kg m}^2$$

[1]

(ii) The centre of mass of a uniform rod is at its midpoint:

$$h = \ell/2 = 0.3 \text{ m}$$

[1]

(iii) Physical pendulum period: (*From formula sheet*)

$$T = 2\pi\sqrt{\frac{I}{Mgh}} = 2\pi\sqrt{\frac{0.096}{0.8 \times 9.81 \times 0.3}} = 2\pi\sqrt{\frac{0.096}{2.354}}$$

$$= 2\pi\sqrt{0.04078} = 2\pi \times 0.2020$$

$$T = 1.27 \text{ s}$$

[1]

Question 4.

[15 marks]

Operators, Eigenvalues, and Expectation Values

- (a) The operator is
- $\hat{A} = 2|1\rangle\langle 1| - |2\rangle\langle 2| + 2|1\rangle\langle 2| + 2|2\rangle\langle 1|$
- .

Taking the Hermitian conjugate: $(|i\rangle\langle j|)^\dagger = |j\rangle\langle i|$, and all coefficients are real. [1]

$$\hat{A}^\dagger = 2|1\rangle\langle 1| - |2\rangle\langle 2| + 2|2\rangle\langle 1| + 2|1\rangle\langle 2| = \hat{A}$$

So $\hat{A}$ is Hermitian. [1]The matrix elements are $A_{ij} = \langle i|\hat{A}|j\rangle$:

$$A_{11} = 2, A_{12} = 2, A_{21} = 2, A_{22} = -1.$$

$$\hat{A} \leftrightarrow \begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix}$$

[1]

- (b) The characteristic equation is
- $\det(\hat{A} - \lambda I) = 0$
- : [1]

$$(2 - \lambda)(-1 - \lambda) - 4 = 0$$

$$-2 - 2\lambda + \lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - \lambda - 6 = 0$$

$$(\lambda - 3)(\lambda + 2) = 0$$

Eigenvalues: $\lambda_+ = 3$ and $\lambda_- = -2$.

[1]

Eigenvector for $\lambda_+ = 3$: $(2 - 3)a + 2b = 0 \Rightarrow -a + 2b = 0 \Rightarrow a = 2b$.Taking $b = 1$, $a = 2$. Normalising: $N = \sqrt{4 + 1} = \sqrt{5}$.

$$|\lambda_+\rangle = \frac{1}{\sqrt{5}}(2|1\rangle + |2\rangle)$$

[1]

Eigenvector for $\lambda_- = -2$: $(2 + 2)a + 2b = 0 \Rightarrow 4a + 2b = 0 \Rightarrow b = -2a$.Taking $a = 1$, $b = -2$. Normalising: $N = \sqrt{1 + 4} = \sqrt{5}$.

$$|\lambda_-\rangle = \frac{1}{\sqrt{5}}(|1\rangle - 2|2\rangle)$$

[1]

- (c) The state is
- $|\psi\rangle = \frac{3}{5}|1\rangle + \frac{4}{5}|2\rangle$
- .

Project onto the eigenstates: (*Recalled — not on formula sheet*)

$$c_+ = \langle \lambda_+ | \psi \rangle = \frac{1}{\sqrt{5}} \left(2 \times \frac{3}{5} + 1 \times \frac{4}{5} \right) = \frac{1}{\sqrt{5}} \cdot \frac{10}{5} = \frac{2}{\sqrt{5}} \quad [1]$$

$$c_- = \langle \lambda_- | \psi \rangle = \frac{1}{\sqrt{5}} \left(1 \times \frac{3}{5} + (-2) \times \frac{4}{5} \right) = \frac{1}{\sqrt{5}} \cdot \frac{-5}{5} = \frac{-1}{\sqrt{5}}$$

Probabilities (Born rule): (*Recalled — not on formula sheet*)

$$P(\lambda_+ = 3) = |c_+|^2 = \frac{4}{5}, \quad P(\lambda_- = -2) = |c_-|^2 = \frac{1}{5} \quad [1]$$

$$\text{Check: } \frac{4}{5} + \frac{1}{5} = 1 \quad \checkmark$$

Possible outcomes: 3 (with probability 4/5) and -2 (with probability 1/5).

[1]

Expectation value: (*Recalled — not on formula sheet*)

$$\langle A \rangle = \lambda_+ P(\lambda_+) + \lambda_- P(\lambda_-) = 3 \times \frac{4}{5} + (-2) \times \frac{1}{5} = \frac{12}{5} - \frac{2}{5} = \frac{10}{5}$$

$$\langle A \rangle = 2$$

[1]

(d) After measuring $\lambda_+ = 3$, the state collapses to: (*Recalled — not on formula sheet*)

$$|\psi(0)\rangle = |\lambda_+\rangle = \frac{1}{\sqrt{5}}(2|1\rangle + |2\rangle)$$

The Hamiltonian is $\hat{H} = E_0 |1\rangle\langle 1| + 2E_0 |2\rangle\langle 2|$, so $|1\rangle$ and $|2\rangle$ are energy eigenstates with energies E_0 and $2E_0$ respectively.

Time evolution: (*From formula sheet*)

$$|\psi(t)\rangle = \frac{1}{\sqrt{5}}(2e^{-iE_0t/\hbar}|1\rangle + e^{-2iE_0t/\hbar}|2\rangle) \quad [1]$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{5}}(2e^{-iE_0t/\hbar}|1\rangle + e^{-2iE_0t/\hbar}|2\rangle)$$

[1]

The probability of measuring $\lambda_+ = 3$ again at time t :

$$P(\lambda_+, t) = |\langle \lambda_+ | \psi(t) \rangle|^2$$

$$\langle \lambda_+ | \psi(t) \rangle = \frac{1}{\sqrt{5}}(2, 1) \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 2e^{-iE_0t/\hbar} \\ e^{-2iE_0t/\hbar} \end{pmatrix} = \frac{1}{5}(4e^{-iE_0t/\hbar} + e^{-2iE_0t/\hbar}) \quad [1]$$

Factor out $e^{-iE_0t/\hbar}$:

$$= \frac{e^{-iE_0t/\hbar}}{5}(4 + e^{-iE_0t/\hbar})$$

$$P(\lambda_+, t) = \frac{1}{25}|4 + e^{-iE_0t/\hbar}|^2 = \frac{1}{25}(16 + 8\cos(E_0t/\hbar) + 1)$$

$$P(\lambda_+ = 3, t) = \frac{17 + 8\cos(E_0t/\hbar)}{25}$$

[1]

Check at $t = 0$: $P = (17 + 8)/25 = 25/25 = 1 \quad \checkmark$ (as expected, since the state was prepared in $|\lambda_+\rangle$).